

Team Worksheet — Lab 25

Response Surface Methodology and Robust Design (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-25-rsm-central-composite.xlsx`.

3. What we produced

Deliverable: An RSM study locating optimal settings plus a robust design analysis against noise factors.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand the limitation of 2-level designs: with only two levels per factor they fit a straight line and cannot detect curvature at all.

Step 2. Add CENTRE POINTS to your factorial design to test for curvature. A significant centre-point effect means the true response is curved and a linear model will mislead you.

Step 3. If no curvature is present and you are far from the optimum, use steepest ascent — move the factor settings in the direction of greatest improvement and re-experiment.

Step 4. Once curvature appears, you are near the optimum and need a second-order design.

Step 5. Learn the central composite design (CCD): a factorial core, plus centre points, plus axial (star) points that add the third level needed to estimate quadratic terms.

Step 6. Learn the Box-Behnken alternative: fewer runs than CCD and it never requires extreme corner combinations, which matters when corners are unsafe or infeasible.

Step 7. Design and run an RSM study on the 2-3 vital factors from Lab 24.

Step 8. Fit the second-order model including linear, quadratic and interaction terms.

Step 9. Generate contour and 3D surface plots to visualise the response surface and locate the optimum region.

Step 10. Find the stationary point and classify it — maximum, minimum or saddle. A saddle point means the optimum lies at a boundary, not in the middle.

Step 11. Use desirability functions if you must optimise several responses at once with competing optima.

Step 12. Now robust design. Separate your factors into CONTROL factors (you set them) and NOISE factors (ambient conditions, material variation, operator, wear).

Step 13. Understand the Taguchi insight: choose control factor settings at which the response is LEAST SENSITIVE to noise — a slightly lower mean with far less variation usually wins.

Step 14. Compute the signal-to-noise ratio for candidate settings, using the correct form for your objective (smaller-is-better, larger-is-better or nominal-is-best).

Step 15. Select the robust operating window and run confirmation trials under deliberately varied noise conditions.

7. Self-check

Curvature is tested with centre points, your second-order model is visualised as a contour or surface plot with a classified stationary point, and the robust settings are confirmed under varied noise conditions.

Complete? ☐ Yes ☐ Not yet

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd